

FIBONACCI ALGORITHM

April 26, 2026

This algorithm is different from the classic Fibonacci algorithm. This new algorithm has a complexity $O(\log n)$. The idea is to calculate the following matrix:

$$\begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}^{n-1}$$

Once the calculation has the result from the top left of this matrix is the corresponding Fibonacci number.

EXAMPLE:

Let's take an example to make it a little clearer:

Let's find the Fibonacci number for $n = 5$. Note $f(n)$ the number fibonacci. Normally the formula Fibonacci classic is $F(n) = F(n-1) + F(n-2)$, where $F(0) = 0$ and $F(1) = 1$. Then $F(5) = 5$. Now in use the matrix, we have

$$\begin{aligned} \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}^4 &= \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}^2 \\ &= \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \\ &= \begin{pmatrix} 3 & 2 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \\ &= \begin{pmatrix} 5 & 3 \\ 3 & 2 \end{pmatrix}. \end{aligned}$$

Hence on the top left, we have $f(5)=5$.